Budget-Resource Labelling: exact pricing for vehicle routing with p-box travel times and unknown dependence

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Abstract

Each carrier in the competing-carrier model solves a vehicle routing problem whose arc travel times are described by p-boxes (lower and upper CDFs) and whose dependence is unknown. The route loss is convex piecewise linear in the route duration. We prove that the worst-case expected loss of a route is the value of a *deterministic* resource allocation problem. Each arc chooses a "deadline budget" d_a and pays a convex cost $c_a(d_a)$, and the budgets of a route must sum to at most the shift length. This is the Meilijson–Nádas stop-loss decomposition, which we re-prove in the form we need. The algorithmic consequence is that the pricing problem of column generation is an elementary shortest path problem with κ additional *continuous decision resources*, where κ is the number of kinks of the loss ($\kappa = 1$ for the loss $\gamma T + \rho(T - T_{\max})^+$). The number of resources does not depend on the resolution of the p-boxes, the sample size or the number of levels L of the quantile grid. Partial paths are compared with a *functional* dominance rule that is valid only on a computable completion window. We prove that the rule is exact and that it subsumes the L-dimensional quantile-vector dominance and the K-dimensional sector dominance. We also prove an envelope version with a label bound per bucket that does not depend on L. Prototype results: the formula agrees with a linear program over all joint laws to $5 \cdot 10^{-14}$; pricing is exact against complete enumeration in 70/70 instances, 18 of them on real metro data; on random instances with $n = 10$–25 customers, BRL prices in 0.004–1.8 s, while quantile-vector, sector-vector and scenario labelling hit 60–120 s limits from $n = 15$; the natural alternative (sweep a single risk level u, solve deterministic problems) is suboptimal in 9/20 small VRPs by up to 6.4%. On the nine metro instances, column generation with BRL pricing converges in 13–680 s and improves on the best heuristic plan in New York (-10%, proved optimal) and Buenos Aires (-1.9%). In three cities it does not beat a heuristic, because price-and-branch is not an exact integer method. Section 9 separates what is known from what is new.

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1 Setting

A *route* r is a depot-to-depot walk; we identify it with the finite multiset of arcs it uses. Arc a has a nonnegative random travel time X_a (service times at nodes can be included as deterministic parts of X_a). The route duration is $T_r = \sum_{a \in r} X_a$.

Assumption 1.1 (p-boxes with bounded support)**.** For every arc a there are CDFs $\underline{F}_a \le \overline{F}_a$ on $[0, B_a]$, $B_a < \infty$ (so $\underline{F}_a(B_a) = 1$). The *credal set* of arc a is $\mathcal{M}_a = \{F \text{ CDF} : \underline{F}_a \le F \le \overline{F}_a\}$. The joint credal set is

$$\mathcal{M} = \{P \text{ law on } \mathbb{R}^A_{\ge 0} : \text{ the } a\text{-th marginal of } P \text{ lies in } \mathcal{M}_a \text{ for every } a\},$$

that is, the marginals are imprecise and the dependence is completely unknown (Fréchet class of a p-box).

The bounded support is needed for $\underline{F}_a$ to be a proper CDF. For a Dvoretzky–Kiefer–Wolfowitz (DKW) band built from n observations with confidence $1-\alpha$, $\underline{F}_a(x) = \max\{F_n(x) - \varepsilon, 0\}$ for $x < B_a$ and $\underline{F}_a(B_a) = 1$, with $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$. For the imprecise Dirichlet model with parameter s, $\underline{F}_a(x) = \#\{i : x_i \le x\}/(n+s)$ for $x < B_a$, again with the remaining mass at B_a.

Definition 1.2 (Upper quantile)**.** $\overline{Q}_a(u) = \inf\{x : \underline{F}_a(x) \ge u\}$ for $u \in (0, 1]$. It is nondecreasing and left-continuous, with $\overline{Q}_a(1) \le B_a$. Write $\overline{Q}_a(0^+) = \lim_{u \downarrow 0} \overline{Q}_a(u)$.

Assumption 1.3 (Loss)**.** The loss of a route of duration T is

$$\varphi(T) = c_0 + \gamma T + \sum_{j=1}^{\kappa} \rho_j \, (T - \theta_j)^+, \qquad \gamma \ge 0,\ \rho_j > 0,\ 0 \le \theta_1 < \cdots < \theta_\kappa.$$

The model of the paper has $\kappa = 1$, $\theta_1 = T_{\max}$, $\rho_1 = \rho$ (Section 2 of `main.tex` and `src/ccu/robust.py`). Every convex nondecreasing φ on a bounded interval is a uniform limit of such functions.

The *robust route cost* is the upper prevision $\overline{\mathbb{E}}[\varphi(T_r)] = \sup_{P \in \mathcal{M}} \mathbb{E}_P[\varphi(T_r)]$. For a loss, this is the Γ-minimax criterion (equivalently Γ-maximin of the payoff $-\varphi$; Lemma 3.5 of `main.tex` for the game-level role). In the game, carrier i minimises $\sum_{r \in \text{plan}} \overline{\mathbb{E}}[\varphi(T_r)] + \lambda^\top x(\text{plan})$, where λ collects the zone prices and congestion charges. Column generation for this problem prices routes with reduced cost $\overline{\mathbb{E}}[\varphi(T_r)] - \sum_{i \in r} \pi_i$, where π_i absorbs the duals and the zone prices of customer i.

Notation: for a CDF F and $d \in \mathbb{R}$, the stop-loss transform is $\pi_F(d) = \mathbb{E}_F \, (X - d)^+$. We write $\overline{\mu}_a = \int_0^1 \overline{Q}_a(u)\, du$ and $\overline{\pi}_a(d) = \mathbb{E}_{\underline{F}_a} \, (X_a - d)^+ = \int_0^1 \left(\overline{Q}_a(u) - d\right)^+ du$. The function $\overline{\pi}_a$ is convex and nonincreasing, has slope -1 for $d < \overline{Q}_a(0^+)$, and vanishes for $d \ge \overline{Q}_a(1)$.

2 The structural theorem

Theorem 2.1 (Upper prevision and budget decomposition)**.** *Under Assumptions 1.1–1.3, for every route r,*

$$\overline{\mathbb{E}}[\varphi(T_r)] = \int_0^1 \varphi\Big(\sum_{a\in r} \overline{Q}_a(u)\Big)\, du \tag{1}$$

$$= c_0 + \sum_{a\in r} \gamma \overline{\mu}_a + \sum_{j=1}^{\kappa} \rho_j \min\Big\{ \sum_{a\in r} \overline{\pi}_a(d_a^j) :\ d^j \in \mathbb{R}^r,\ \sum_{a\in r} d_a^j \le \theta_j \Big\}. \tag{2}$$

The supremum over $\mathcal{M}$ is attained by the comonotone coupling of the lower CDFs, $X_a = \overline{Q}_a(U)$ with U uniform. Each minimum in (2) *is attained. If $\{\overline{Q}_a(u)\}$ and θ_j are integers, it is attained at an integer vector d^j.*

Proof. Write $S = \sum_{a\in r} X_a$ (arcs used several times are counted with multiplicity; the argument below never needs distinct arcs). Fix j and $\theta = \theta_j$.

Step 1 (pointwise inequality). For reals x_a and d_a with $\sum_a d_a \le \theta$, $\big(\sum_a x_a - \theta\big)^+ \le \big(\sum_a (x_a - d_a)\big)^+ \le \sum_a (x_a - d_a)^+$. The first inequality holds because $t \mapsto t^+$ is nondecreasing; the second because $t \mapsto t^+$ is subadditive.

Step 2 (upper bound). Let $P \in \mathcal{M}$ with marginals $F_a \in \mathcal{M}_a$. Taking expectations in Step 1, $\mathbb{E}_P\,(S-\theta)^+ \le \sum_a \pi_{F_a}(d_a)$. Since $F_a \ge \underline{F}_a$ pointwise, $X_a \sim F_a$ is first-order stochastically dominated by $\underline{F}_a$, and $x \mapsto (x-d)^+$ is nondecreasing, so $\pi_{F_a}(d_a) \le \overline{\pi}_a(d_a)$. In the same way, $\mathbb{E}_{F_a} X_a \le \overline{\mu}_a$. Hence for every $P \in \mathcal{M}$ and every admissible $d^1, \dots, d^\kappa$,

$$\mathbb{E}_P \varphi(S) \le c_0 + \gamma \sum_a \overline{\mu}_a + \sum_j \rho_j \sum_a \overline{\pi}_a(d_a^j).$$

So $\overline{\mathbb{E}}[\varphi(S)]$ is at most the right-hand side of (2).

Step 3 (attainment). Let U be uniform on $(0,1)$, $\bar{X}_a = \overline{Q}_a(U)$ and $\bar{S} = \sum_a \bar{X}_a = \Sigma(U)$ with $\Sigma(u) = \sum_a \overline{Q}_a(u)$. Then $\bar{X}_a \sim \underline{F}_a \in \mathcal{M}_a$, so the law $\bar{P}$ of $(\bar{X}_a)$ belongs to $\mathcal{M}$. Fix j, $\theta = \theta_j$. Let $u^* = \sup\{u \in (0,1] : \Sigma(u) \le \theta\}$, with $\sup \emptyset = 0$. If $u^* = 1$, then $\bar{S} \le \theta$ almost surely; put $d_a = \overline{Q}_a(1)$. Then $\sum_a d_a = \Sigma(1) \le \theta$ (left-continuity), and both sides of Step 1 vanish almost surely. If $u^* < 1$, put $\ell_a = \overline{Q}_a(u^*)$ (with $\ell_a = \overline{Q}_a(0^+)$ if $u^* = 0$) and $h_a = \overline{Q}_a(u^{*+})$. By the definition of u^* and left-continuity, $\sum_a \ell_a \le \theta$ when $u^* > 0$, and $\sum_a h_a \ge \theta$. If $u^* = 0$, choose any $d_a \le \ell_a$ with $\sum_a d_a = \theta$ (possible because $\sum_a \ell_a = \Sigma(0^+) \ge \theta$). Otherwise choose $d_a \in [\ell_a, h_a]$ with $\sum_a d_a = \theta$, which is possible by the intermediate value theorem. Then $\{U \le u^*\}$ implies $\bar{X}_a \le \ell_a \le d_a$ for all a, and $\{U > u^*\}$ implies $\bar{X}_a \ge h_a \ge d_a$ for all a. On the first event both sides of Step 1 are 0; on the second, $\sum_a \big(\bar{X}_a - d_a\big)^+ = \sum_a (\bar{X}_a - d_a) = \bar{S} - \theta = \big(\bar{S} - \theta\big)^+$, since the sum is nonnegative. (For $u^* = 0$ the first event is null.) So Step 1 holds with equality $\bar{P}$-almost surely for this d. Taking expectations, $\mathbb{E}_{\bar{P}}\,\big(\bar{S} - \theta\big)^+ = \sum_a \overline{\pi}_a(d_a)$. Doing this for each j and adding $\mathbb{E}\bar{S} = \sum_a \overline{\mu}_a$ shows that $\bar{P}$ attains the bound of Step 2 and that the minima in (2) are attained.

Step 4 (formula (1)). $\mathbb{E}_{\bar{P}} \varphi(\bar{S}) = \int_0^1 \varphi(\Sigma(u)) du$ because $\bar{S} = \Sigma(U)$.

Step 5 (integrality). If all $\overline{Q}_a$ take integer values and θ is an integer, then ℓ_a, h_a and $\overline{Q}_a(1)$ in Step 3 are integers. In each of the three cases an integer vector d with the stated properties exists: $d = \overline{Q}(1)$; an integer vector with $d \le \ell$ and $\sum d = \theta$ (when $\sum \ell \ge \theta$); or an integer vector in the box $[\ell, h]$ with sum θ (when $\sum \ell \le \theta \le \sum h$). The equality argument of Step 3 applies verbatim. □

Remark 2.2 (What is classical)**.** Equation (1) and the identity $\mathbb{E}\,(S^c - \theta)^+ = \min_{\sum d_a = \theta} \sum_a \pi_a(d_a)$ for a comonotone sum with fixed marginals are due to Meilijson and Nádas [1]. They also appear in

actuarial science as the stop-loss decomposition of comonotone sums [2, 3]. The p-box step, which picks the lower CDF by first-order dominance, is standard in probability-bound analysis [4, 5]. We include the proof because the *routing* statements below depend on the precise form (2) with *inequality* budgets, on Step 3's choice of d, and on integrality.

Corollary 2.3 (Common level grid)**.** *If all $\overline{Q}_a$ are step functions constant on the intervals $(u_{l-1}, u_l]$, $l = 1, \dots, L$, with $w_l = u_l - u_{l-1}$ and values $q_{a,l}$, then with $S_l = \sum_{a \in r} q_{a,l}$,*

$$\overline{\mathbb{E}}[\varphi(T_r)] = \sum_{l=1}^{L} w_l \varphi(S_l), \qquad f_r(D) := \gamma \sum_l w_l S_l + \rho \sum_l w_l \left(S_l - D\right)^+ .$$

For $\kappa = 1$, $\overline{\mathbb{E}}[\varphi(T_r)] = c_0 + f_r(T_{\max})$. DKW and IDM p-boxes built from samples of a common size share a grid. In general, the union of the breakpoints is a common grid with $L \le \sum_a (\#\text{breakpoints of } \overline{Q}_a)$.

Remark 2.4 (Lower previsions are different)**.** The *lower* prevision $\inf_{P \in \mathcal{M}} \mathbb{E}_P \varphi(S)$ is not given by a comonotone or any other universal coupling for three or more arcs; it is a joint-mix problem. Theorem 2.1 is a statement about the upper prevision of a convex loss only.

2.1 Consequence for routing: decision resources

Theorem 2.5 (Budget-resource reformulation)**.** *Let $\mathcal{R}$ be the set of feasible routes (capacity, elementarity) and $\pi \in \mathbb{R}^n$ dual prices. With $c_a(d^1, \dots, d^\kappa) = \gamma \overline{\mu}_a + \sum_j \rho_j \overline{\pi}_a(d^j)$, the dependence-robust pricing problem satisfies*

$$\min_{r \in \mathcal{R}} \Big\{ \overline{\mathbb{E}}[\varphi(T_r)] - \sum_{i \in r} \pi_i \Big\} = \min \Big\{ c_0 + \sum_{a \in r} c_a(d_a^1, \dots, d_a^\kappa) - \sum_{i \in r} \pi_i : \ r \in \mathcal{R}, \ \sum_{a \in r} d_a^j \le \theta_j \ (j \le \kappa) \Big\}.$$

This is a deterministic elementary shortest path problem with κ additional resources. The consumption d_a^j of resource j on arc a is a decision, paid through the separable convex arc cost c_a. The number of resources is κ and does not depend on the p-box resolution L, on the sample size, or on the number of arcs. The same holds for the full routing problem: a set-partitioning (or arc-flow) model with route costs $\overline{\mathbb{E}}[\varphi(T_r)]$ equals the model in which each vehicle carries κ budget resources.

Proof. Substitute (2) into the objective and merge the two minimisations. □

Remark 2.6 (Scenario view and why it is worse)**.** Corollary 2.3 also shows that the robust problem is a scenario problem with L comonotone scenarios $q_{\cdot,l}$ and probabilities w_l. This is the reduction of Ghosal, Ho and Wiesemann for demand ambiguity in the CVRP [6]. A scenario-based labelling algorithm then needs L resources (the vector $(S_l)_l$) with pointwise dominance. Theorem 2.5 replaces these L resources by κ decision resources. The rest of the paper turns this into a dominance rule.

3 Labels, completion windows and functional dominance

We work with $\kappa = 1$ ($\theta = T_{\max}$) and the common grid of Corollary 2.3. The extension to $\kappa > 1$ is in Remark 3.8. A *label* is a depot-to-v path p. It carries its end node v, visited set $V(p)$, load $\ell(p)$, level profile $S_p \in \mathbb{R}^L$ (nondecreasing in l), and collected duals $\pi(p)$. Its *cost function* is

$$f_p(D) = \gamma \sum_l w_l S_{p,l} + \rho \sum_l w_l \left(S_{p,l} - D\right)^+ , \qquad D \in \mathbb{R}.$$

It is convex, nonincreasing and piecewise linear, with breakpoints at the values $S_{p,l}$, slope $-\rho$ for $D < S_{p,1}$, and constant for $D \geq S_{p,L}$. Write $S_p(0^+) = S_{p,1}$ and $S_p(1) = S_{p,L}$.

A *completion* of p is a path q from v to the depot such that $p \oplus q \in \mathcal{R}$. Its cost function f_q is defined in the same way.

Lemma 3.1 (Concatenation is inf-convolution)**.** *For a label p and a completion q,*

$$\overline{\mathbb{E}}[\varphi(T_{p\oplus q})] - c_0 = f_{p\oplus q}(\theta) = \min_{D\in\mathbb{R}} \big\{ f_p(D) + f_q(\theta - D) \big\} = (f_p \,\square\, f_q)(\theta).$$

Proof. By (2), $\overline{\pi}$-terms are separable over arcs. Split the budget vector into the arcs of p and those of q, put $D = \sum_{a\in p} d_a$, and minimise first over the arcs of p with $\sum_{a\in p} d_a \leq D$ and then over those of q with $\sum_{a\in q} d_a \leq \theta - D$. Applying (2) to p and q separately (with θ replaced by D and $\theta - D$) gives $f_p(D)$ and $f_q(\theta - D)$. The mean terms add. □

Lemma 3.2 (Completion window)**.** *Let $C_{\min}(v) \leq S_q(0^+)$ for every completion q of labels ending at v; for example, $C_{\min}(v)$ is the shortest-path distance from v to the depot with arc lengths $\overline{Q}_a(0^+)$. Put*

$$h(p) = \min\{S_p(1),\ \theta - C_{\min}(v)\}, \qquad \lambda(p) = \min\{S_p(0^+), h(p)\}, \qquad J(p) = [\lambda(p), h(p)].$$

For every completion q, the minimum in Lemma 3.1 is attained at some $D^* \leq h(p)$. If $D^* < \lambda(p)$, then $D^* \leq S_p(0^+)$.

Proof. Let $\Phi(D) = f_p(D) + f_q(\theta - D)$. The function $D \mapsto f_q(\theta - D)$ is convex, nondecreasing, with slope in $[0, \rho]$, and slope exactly ρ when $\theta - D < S_q(0^+)$. The slope of f_p lies in $[-\rho, 0]$ and is 0 for $D > S_p(1)$. (i) On $D > \theta - S_q(0^+)$, Φ has slope $\geq -\rho + \rho = 0$, so it is nondecreasing there and a minimiser exists with $D \leq \theta - S_q(0^+) \leq \theta - C_{\min}(v)$. (ii) On $D > S_p(1)$, Φ has slope ≥ 0, so decreasing D to $S_p(1)$ does not increase Φ. Applying (i) and then (ii) gives a minimiser with $D^* \leq h(p)$. The last claim is trivial: if $D^* < \lambda(p) = \min\{S_p(0^+), h(p)\}$, then $D^* < S_p(0^+)$. □

Definition 3.3 (Functional dominance)**.** Label p_1 *functionally dominates* p_2, both ending at v, if $V(p_1) \subseteq V(p_2)$, $\ell(p_1) \leq \ell(p_2)$, and

$$g(D) := \big(f_{p_1}(D) - \pi(p_1)\big) - \big(f_{p_2}(D) - \pi(p_2)\big) \leq 0 \qquad \text{for all } D \in J(p_2).$$

Since g is piecewise linear with breakpoints in $\{S_{p_1,l}\} \cup \{S_{p_2,l}\}$, it suffices to check the two endpoints of $J(p_2)$ and the breakpoints inside it: at most $2L + 2$ evaluations, each $O(\log L)$ with suffix sums.

Definition 3.4 (Completion dominance)**.** p_1 *completion-dominates* p_2 if every completion q of p_2 is a completion of p_1 and $\overline{\mathbb{E}}[\varphi(T_{p_1\oplus q})] - \pi(p_1 \oplus q) \leq \overline{\mathbb{E}}[\varphi(T_{p_2\oplus q})] - \pi(p_2 \oplus q)$. Completion dominance is a preorder (reflexive and transitive).

Theorem 3.5 (Validity of functional dominance)**.** *Functional dominance implies completion dominance.*

Proof. Let q be a completion of p_2. Since $V(p_1) \subseteq V(p_2)$ and $\ell(p_1) \leq \ell(p_2)$, q is a completion of p_1 (elementarity and capacity). The duals of q appear on both sides. By Lemma 3.2 applied to p_2, there is a minimiser D^* for $p_2 \oplus q$ with $D^* \leq h(p_2)$. If $D^* \in J(p_2)$, then $g(D^*) \leq 0$. Otherwise $D^* < \lambda(p_2) \leq S_{p_2}(0^+)$. On $(-\infty, S_{p_2}(0^+)]$, f_{p_2} has slope $-\rho$ and f_{p_1} has slope $\geq -\rho$, so g is nondecreasing there. Since $\lambda(p_2) \leq S_{p_2}(0^+)$, $g(D^*) \leq g(\lambda(p_2)) \leq 0$. In both cases, by Lemma 3.1,

$$f_{p_1\oplus q}(\theta) - \pi(p_1) \leq f_{p_1}(D^*) + f_q(\theta - D^*) - \pi(p_1) \leq f_{p_2}(D^*) + f_q(\theta - D^*) - \pi(p_2) = f_{p_2\oplus q}(\theta) - \pi(p_2). \ \square$$

Proposition 3.6 (Comparison with vector dominance)**.** *Assume $V(p_1) \subseteq V(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $\pi(p_1) \ge \pi(p_2)$.*

(a) (Quantile-vector dominance, L resources.) If $S_{p_1} \le S_{p_2}$ componentwise, then p_1 functionally dominates p_2.

(b) (Sector dominance, K resources.) In the sector model $X_a = \sum_k \tau_{a,k}\xi_k$ with p-boxes on the multipliers ξ_k, so that $S_{p,l} = \sum_k \tau_{p,k}\overline{Q}_{\xi_k}(u_l)$ + (service): if $\tau_{p_1} \le \tau_{p_2}$ componentwise, then $S_{p_1} \le S_{p_2}$, hence (a) applies.

(c) Neither converse holds. There are pairs with $f_{p_1} \le f_{p_2}$ on $\mathbb{R}$ and $S_{p_1,l} > S_{p_2,l}$ for some l.

Proof. (a) f is nondecreasing in each S_l. (b) The $\overline{Q}_{\xi_k}$ are nonnegative and the service part is monotone in V. (c) Take $L = 2$, $w = (\frac{1}{2}, \frac{1}{2})$, $\gamma = 0$, $\rho = 1$ and equal duals. *First example, whole line:* $S_{p_1} = (1, 1)$ and $S_{p_2} = (0, 2.2)$. Then $f_{p_1}(D) = (1 - D)^+$ and $f_{p_2}(D) = \frac{1}{2}[(-D)^+ + (2.2 - D)^+]$. At the breakpoints $D = 0, 1, 2.2$ the values are $(1, 1.1)$, $(0, 0.6)$ and $(0, 0)$; for $D \le 0$ the difference is the constant -0.1, and for $D \ge 2.2$ both vanish. Hence $f_{p_1} \le f_{p_2}$ on $\mathbb{R}$, while $S_{p_1,1} = 1 > 0 = S_{p_2,1}$. *Second example, the window matters:* $S_{p_1} = (0.5, 2.5)$ and $S_{p_2} = (1.5, 1.6)$, with $\theta - C_{\min}(v) = 0.5$. Then $J(p_2) = \{0.5\}$, $f_{p_1}(0.5) = 1 < 1.05 = f_{p_2}(0.5)$, so p_2 is pruned. On the whole line, however, $f_{p_1}(2) = 0.25 > 0 = f_{p_2}(2)$, and $S_{p_1,2} > S_{p_2,2}$. □

Remark 3.7. Proposition 3.6(c) shows where the extra pruning comes from. On the window, the comparison only involves the part of the stop-loss transform that a completion can still reach. When $\theta - C_{\min}(v) \le S_{p_2}(0^+)$ (the route is late at every level whatever happens next), the window is the single point $\lambda = \theta - C_{\min}(v)$ and the comparison is one scalar inequality, $f_{p_1}(\lambda) - \pi(p_1) \le (\gamma + \rho)\overline{\mu}_{p_2} - \rho\lambda - \pi(p_2)$. When $\theta - C_{\min}(v) \ge S_{p_2}(1)$, the window is $[S_{p_2}(0^+), S_{p_2}(1)]$, and the rule is the stop-loss (increasing-convex) order of the two worst-case route durations, shifted by the duals.

Remark 3.8 (Several kinks)**.** For $\kappa > 1$, a label's cost function is $f_p(D^1, \dots, D^\kappa) = \gamma\overline{\mu}_p + \sum_j \rho_j \pi_{S_p}(D^j)$, which is separable in j. Lemma 3.1 holds coordinatewise and Lemma 3.2 holds for each j with θ_j. A sufficient dominance condition is $\gamma(\overline{\mu}_{p_1} - \overline{\mu}_{p_2}) + \sum_j \rho_j \max_{D \in J_j(p_2)}[\pi_{S_{p_1}} - \pi_{S_{p_2}}](D) \le \pi(p_1) - \pi(p_2)$, where the mean term is separated out of each f. Each maximum is over a piecewise linear function and is computed at breakpoints; on the part of the window below $S_{p_2}(0^+)$ the same monotonicity argument applies to each term.

4 The algorithm BRL

Theorem 4.1 (Exactness of BRL)**.** *BRL returns $\min_{r \in \mathcal{R}}\{\overline{\mathbb{E}}[\varphi(T_r)] - \pi(r)\}$ exactly, with a route attaining it.*

Proof. Call a label *processed* if it is popped while not deleted; processed labels are extended and closed. Let OPT be the optimum and let Ω be the set of pairs (o, q) with o a created label, q a completion of o and $\overline{\mathbb{E}}[\varphi(T_{o \oplus q})] - \pi(o \oplus q) = \text{OPT}$. The pair (depot label, optimal route) is in Ω, and the depot label is processed. Among pairs in Ω with o processed, choose one with the smallest number of arcs $|q|$. If $|q| = 1$, then q is the return arc and closing o records a route of value OPT. If $|q| \ge 2$, let (v, j) be the first arc of q and q' the rest, and let $o_1 = o \oplus (v, j)$, which is created when o is processed; $(o_1, q') \in \Omega$. If o_1 is processed, $|q'| < |q|$ contradicts minimality. Otherwise o_1 was discarded at creation or deleted later, in both cases because a label o_2, not deleted at that moment, functionally dominates it. By Theorem 3.5, o_2 completion-dominates o_1, so $(o_2, q') \in \Omega$. If o_2 is not processed, it was deleted at a later moment by a label o_3, and transitivity of completion dominance

Algorithm 1 BRL: Budget-Resource Labelling (pricing for the dependence-robust VRP)

Require: graph, demands, capacity Q; level grid (w_l); arc level profiles $q_{a,\cdot}$ (Cor. 2.3); $\gamma, \rho, \theta, c_0$; duals π

$C_{\min}(v) \leftarrow$ shortest path $v \to$ depot with lengths $q_{a,1}$ ▷ Lemma 3.2
$\mathcal{B}_v \leftarrow \emptyset$ for all v; queue $\leftarrow \{(\text{depot}, \emptyset, 0, 0 \in \mathbb{R}^L, 0)\}$
while queue nonempty **do**
  pop p (FIFO by number of visited customers)
  close: $rc \leftarrow c_0 + f_{p\oplus(v,0)}(\theta) - \pi(p)$; record if $rc < 0$
  for each customer $j \notin V(p)$ with $\ell(p) + d_j \le Q$ **do**
    $p' \leftarrow (j, V(p) \cup \{j\}, \ell(p) + d_j, S_p + q_{(v,j),\cdot}, \pi(p) + \pi_j)$; precompute suffix sums of $w \odot S_{p'}$
    if some $o \in \mathcal{B}_j$ functionally dominates p' **then** discard p' ▷ Def. 3.3
    else delete from $\mathcal{B}_j$ the labels functionally dominated by p'; insert p'; push p'
    end if
  end for
end while
return the recorded routes sorted by rc and $\min rc$

gives $(o_3, q') \in \Omega$. The moments at which $o_1, o_2, \ldots$ cease to be live strictly increase and there are finitely many labels, so the chain ends at a processed label o_k with $(o_k, q') \in \Omega$ and $|q'| < |q|$. This contradicts minimality, so the case $|q| \ge 2$ is impossible. □

Column generation (the method on real instances). Master: $\min\{\sum_r c_r x_r : \sum_{r\ni i} x_r \ge 1,\ x \ge 0\}$ with exact $c_r = \overline{\mathbb{E}}[\varphi(T_r)]$ from Corollary 2.3. Pricing is BRL. When BRL returns $\min rc \ge 0$, the master value is the exact LP bound of the dependence-robust VRP. Integer solutions come from the master restricted to the generated columns (price-and-branch) or from branching on arcs, as usual.

ng-routes. On the metro instances we price over ng-routes (memory sets in place of visited sets). Theorem 3.5 holds verbatim with $V(p)$ replaced by the ng-memory, because a completion that is ng-feasible for p_2 is ng-feasible for any p_1 whose memory is a subset. The bound is the (weaker, valid) ng-relaxation bound.

Add-on: certified gap. Because BRL is exact, the final master value is a valid lower bound, and any integer solution gives a gap that holds for the robust objective itself. The common-level alternative (Section 5) gives only a weaker bound.

5 Why one risk level is not enough

A natural shortcut uses the dual form $\mathbb{E}\,(S-\theta)^+ = \max_{u\in[0,1]} \big[\int_u^1 \Sigma(s)ds - (1-u)\theta\big]$, where Σ is the quantile of S. This gives

$$\overline{\mathbb{E}}[\varphi(T_r)] = c_0 + \gamma\overline{\mu}_r + \rho \max_{u\in[0,1]} \Big[\sum_{a\in r} H_a(u) - (1-u)\theta\Big], \qquad H_a(u) = \int_u^1 \overline{Q}_a(s)ds.$$

Fixing one u gives additive arc costs $\gamma\overline{\mu}_a + \rho H_a(u)$, which can be fed to a deterministic VRP. Sweeping u over the grid and evaluating each solution exactly gives a heuristic and the bound $\max_u \min_{\text{plans}}$.

Proposition 5.1. *(a) For every u, the fixed-u problem is a lower bound:* $\min_{plan} \sum_r \overline{\mathbb{E}}[\varphi(T_r)] \geq \max_u \min_{plan} \sum_r [c_0 + \gamma \overline{\mu}_r + \rho(H_r(u) - (1-u)\theta)]$. *(b) The inequality can be strict, and the best fixed-u plan need not be optimal.*

Proof. (a) $\max_u$ inside each route is at least its value at a common u, and $\min \max \geq \max \min$. (b) The numerical counterexamples in `results.md` (Test F: 9 of 20 instances, gap up to 6.4%) are exact computations by enumeration and set partitioning. Structurally, the optimal u of a route is the worst-case on-time level $u_r^* = \sup\{u : \Sigma_r(u) \leq \theta\}$, which differs between routes of the same plan. A single u cannot match all of them. □

The budget resource of Theorem 2.5 is exactly what lets each route choose its own level inside the pricing problem.

6 Envelope dominance and a label bound independent of L

Proposition 6.1 (Envelope dominance)**.** *Assume integer data: all $q_{a,l}$ and θ are integer multiples of a time unit δ, measured in units of δ. Let p and $p_1, \ldots, p_m$ end at v, with $V(p_i) \subseteq V(p)$ and $\ell(p_i) \leq \ell(p)$. If*

$$f_p(D) - \pi(p) \;\geq\; \min_i \big(f_{p_i}(D) - \pi(p_i)\big) \quad \text{for every integer } D \in J(p),$$

then for every completion q of p some $p_i \oplus q$ has reduced cost at most that of $p \oplus q$. Consequently, after envelope pruning, a bucket (end node, visited set, load) holds at most $|J \cap \mathbb{Z}| \leq \theta/\delta + 1$ labels, for any L.

Proof. f_p and f_q are piecewise linear with integer breakpoints, so $\Phi(D) = f_p(D) + f_q(\theta - D)$ is too. A minimiser can be taken at an integer point; Lemma 3.2 restricts it to $D^* \leq h(p)$, where $h(p)$ is an integer, and the moves in its proof keep integrality. If $D^* \in J(p)$, apply the hypothesis at D^* and pick the minimising i. If $D^* < \lambda(p)$, every $g_i = (f_{p_i} - \pi(p_i)) - (f_p - \pi(p))$ is nondecreasing on $(-\infty, S_p(0^+)]$ (proof of Theorem 3.5). So $\min_i g_i(D^*) \leq \min_i g_i(\lambda(p)) \leq 0$, and $\lambda(p)$ is an integer. For the bound, each surviving label must be the strict minimiser (after fixed tie-breaking) of $f - \pi$ at some integer of its window. All labels of a bucket share v. If $\theta - C_{\min}(v) \geq 0$ their windows lie in $[0, \theta]$, which contains at most $\theta/\delta + 1$ integers. Otherwise every window is the single common point $\theta - C_{\min}(v)$ and at most one label survives. □

Remark 6.2. Pairwise dominance has no such bound: with L free levels, f-functions can cross at many points. The prototype implements the pairwise rule only (Definition 3.3), because it already keeps few labels (Section 8). Envelope dominance is stated for completeness and is **not** part of the tested method.

7 Zones and a common traffic factor

Arcs in different zones share a city-wide traffic factor G. Suppose G takes finitely many values $g \in \mathcal{G}$ with probabilities p_g, that given $G = g$ arc a has a p-box $[\underline{F}_{a|g}, \overline{F}_{a|g}]$, and that the dependence between arcs given G is unknown. Let $\mathcal{M}^G$ be the corresponding set of joint laws of (G, X).

Theorem 7.1 (Conditionally comonotone upper prevision)**.** $\sup_{P \in \mathcal{M}^G} \mathbb{E}_P \varphi(T_r) = \sum_g p_g \int_0^1 \varphi\big(\sum_{a \in r} \overline{Q}_{a|g}(u)\big) du$. *The robust pricing problem is an elementary shortest path problem with $\kappa|\mathcal{G}|$ budget resources, and functional dominance applies atom by atom. If the law of G is itself only known to lie in a p-box, and the conditional p-boxes increase with g in first-order dominance, the worst law of G is its lower CDF and the same formula holds with that law.*

Proof. Conditioning on G splits $\mathcal{M}^G$ into a product over atoms of Fréchet classes, because a joint law is any mixture $\sum_g p_g P_g$ with P_g in the conditional class. The supremum of a sum of terms that depend on separate variables P_g is the sum of the suprema; apply Theorem 2.1 to each atom. The resource statement follows by applying Theorem 2.5 per atom. For the last claim, $g \mapsto \int_0^1 \varphi(\sum_a \overline{Q}_{a|g})$ is nondecreasing when the $\overline{Q}_{a|g}$ increase in g, and a nondecreasing function has its largest expectation under the stochastically largest law in the p-box of G. □

Remark 7.2 (Hierarchy of resource counts)**.** Unknown dependence needs κ resources. Conditional comonotonicity given an m-atom factor needs κm. A full empirical joint law with S scenarios (the precise-probability sample-average model) needs S scenario resources. Under independence with exact distributions, evaluating one route is already #P-hard (counting knapsack solutions), so no finite resource description of fixed size exists unless #P collapses. In the engine's sector model, $X_a = \sum_k \tau_{a,k}\xi_k$, arcs in the same sector are comonotone by construction, so Theorem 2.1 applies with p-boxes on the ξ_k. Its value is also the worst case over the restricted class in which within-sector dependence is fixed, because the comonotone coupling belongs to that class.

8 Numerical evidence (summary)

Full tables are in `results.md`; code is in `prototype.py` and `real_instances.py`.

- **Formula vs brute force.** 40 random p-box instances with 2–3 arcs: an LP over all joint pmfs on the product grid agrees with (1) to $4.5 \cdot 10^{-14}$ relative error. The factor model of Theorem 7.1 (20 instances) agrees to $2.5 \cdot 10^{-14}$.
- **Budget identity.** 60 instances: the budget LP and the integer DP agree with (1) to $1.4 \cdot 10^{-14}$.
- **Exactness of BRL.** 52 random ESPPRC instances ($n \le 9$, deadline fraction 0.3–2.0, $K \in \{1, 4, 9\}$) and 18 real metro sub-instances (nine cities, 9 customer zones, column-generation duals scaled by 1 and 1.3): the minimum reduced cost of BRL equals complete enumeration in all 70.
- **Pruning and scaling.** Random instances, $K = 9$, $L = 33$, capacity 6. For $n = 10, 15, 20, 25$, BRL takes 0.004–0.065 s (16–109 labels). Sector dominance times out at 60 s from $n = 10$; quantile-vector dominance and the 40-scenario SAA labelling time out from $n = 15$. With duals scaled by 2.5, BRL takes ≤ 1.8 s up to $n = 25$, while even the risk-neutral ESPPRC times out at $n = 25$. With loose deadlines (1.5–5×), BRL takes ≤ 2 s.
- **One level is not enough.** Test F: the fixed-u sweep is suboptimal in 9/20 instances (up to 6.4%), and its bound gap reaches 22% (median 6.9%).
- **Metro instances.** 22 customer zones per city, $L = 51$, ng-routes with $|N| = 8$. Column generation converges with exact pricing in all nine cities (13–680 s). The LP-to-plan gaps are 0–4%. New York's plan is proved optimal and is 10% below every heuristic; Buenos Aires improves by 1.9%. In Berlin, London and Paris an SAA-optimised heuristic plan is 0.1–1.3% better under the p-box objective than our price-and-branch plan, because branching is not implemented.
- **Model conservatism (not an algorithmic issue).** On the metro instances the p-box worst case is 21–36% above the expected cost under the calibrated true law. Unknown dependence accounts for 5–7% of this; the rest comes from the DKW marginals and the support cap.

9 Prior art: what is known and what is new

Known. Equation (1) and the stop-loss decomposition (2) for fixed marginals [1, 2, 3]. Worst-case marginals of a p-box under monotone losses [4, 5]. The comonotone worst-case reduction of a CVRP with ambiguous demand marginals to a scenario CVRP solved by branch-and-cut [6]. Label-setting ESPPRC and resource extension functions [7, 8]. Second-order stochastic dominance between paths for *independent* arc times [9]. Routing with lateness indices under independence [10] or moment ambiguity [11, 12]. Worst-case CVaR additivity under unknown dependence. Distributionally robust project crashing without correlation information [13].

New, as far as our searches show (snippet-level; see `algorithm.md`). (i) Using the stop-loss decomposition to turn the per-arc budget into a *decision resource* of the pricing problem (Theorem 2.5). The pricing then has κ resources instead of L scenario resources. (ii) The completion-window functional dominance (Lemma 3.2, Theorem 3.5) and its exactness proof. The window argument uses the slope bounds $[-\rho, 0]$, which are specific to the inf-convolution structure of comonotone sums. Under independence, concatenation is a convolution of laws, not an inf-convolution of stop-loss transforms, and Lemma 3.1 fails. (iii) The factor extension with $\kappa|\mathcal{G}|$ resources (Theorem 7.1). (iv) The evidence that the common-level shortcut is not exact (Proposition 5.1). **Not claimed**: polynomial solvability (the pricing problem is NP-hard, since it contains the deterministic ESPPRC), novelty of (1), and envelope dominance in the tested method.

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